

Yucheng (Cecilia) Zhang

737-343-4836 | ceciliayczhang@gmail.com | Portfolio | LinkedIn | GitHub

EDUCATION

Northwestern University

M.S. in Computer Science

Aug. 2026 – May 2028

Evanston, IL

The University of Texas at Austin

B.S. in Mathematics, Minor in Computer Science | GPA: 3.74/4.00

Aug. 2023 – May 2026

Austin, TX

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, R

Backend/Web: React, Flask, Node.js, REST APIs, SQLAlchemy, Alembic, Pytest, Jest

Data/Cloud: PostgreSQL, SQL Server, Azure Synapse, Databricks, Docker, Git, Power BI

AI Integrations: OpenAI, Gemini, Claude, LLM APIs, prompt templates, context filtering, AI feedback systems

SOFTWARE ENGINEERING EXPERIENCE

CodeAssist – Software Engineer / Researcher | GitHub

Aug. 2025 – Present

University of Texas at Austin | React, Flask, PostgreSQL, Docker

Austin, TX

- Built assignment-level AI feedback controls, enabling instructors to configure prompt templates, approved inputs, model settings, request limits, and cooldown rules per assignment.
- Developed Flask REST APIs with backend validation, JSON serialization, SQLAlchemy models, Alembic migrations, and instructor/TA authorization safeguards.
- Implemented LLM context-filtering logic so only instructor-approved assignment descriptions, student code, tests, results, and output are sent to AI providers.
- Improved multi-provider workflows across OpenAI, Gemini, Claude, and local models, including model selection, timeout handling, API configuration, and settings consistency fixes.
- Created backend stress-testing tools with safe local runners, health checks, structured JSON reports, cleanup options, and documentation for enrollment, submissions, grading, and CRUD workflows.

Enterprise Risk Analyst Intern

May 2025 – Dec. 2025

Toyota Industries Commercial Finance | Python, SQL, Azure, Power BI

Dallas, TX

- Automated monthly credit-risk model refreshes with parameterized SQL and Python workflows connected to Power BI, reducing manual reporting effort by 10%.
- Built ETL and reconciliation logic across PayNet and internal loan servicing data, standardizing features including inquiries, LTV ratio, and account tenure.
- Delivered Azure and Power BI dashboards for real-time risk tracking across Class 8 trucks, trailers, and used-vehicle loan portfolios.
- Validated Logistic Regression and Random Forest outputs with ROC/AUC, confusion matrices, and feature importance to support scorecard refinements.

Software Engineering Intern – Customer Analytics

Nov. 2023 – Jun. 2024

Joblogic-X Corporation | Python, APIs, Git

Dallas, TX

- Engineered a pipeline to scrape and process XHS social-media data and integrate it with transactional order systems for scalable storage and analysis.
- Implemented modular Python ranking and clustering algorithms, generating reproducible customer-priority lists for marketing and engagement workflows.
- Integrated a KOC scoring module into Meet Fresh's AI-powered chatbot through REST APIs, improving personalization and product usability.
- Maintained Git-based deployment workflows to keep the codebase reproducible, cross-platform, and ready for future extensions.

Financial Data Development Engineer – Fraud Detection

Jul. 2024 – Aug. 2024

Caitong Securities Ltd | SQL, Oracle, Stored Procedures

Hangzhou, China

- Designed SQL ETL pipelines processing 10M+ daily transactions with window functions, subqueries, and stored procedures, improving extraction efficiency by 5%.
- Built modular fraud-detection logic and a customer risk-scoring engine from transactional and behavioral streams, reducing false positives by 2%.

SELECTED PROJECT

Echo Prototype – Interactive Voice Simulation

2026

Next.js, TypeScript, OpenAI API, Browser TTS

Personal Project

- Built an event-driven web app that generates context-aware voice responses from user actions, pauses, errors, and task switching; implemented session state, trigger rules, audio cooldowns, and API-based generation.
- Implemented front-end session controls, asynchronous response handling, and browser TTS playback so generated feedback stays responsive, non-repetitive, and tied to user behavior.
- Structured reusable prompt and event logic for task switching, error states, and idle moments, creating a foundation for future AI assistant prototypes.

Data Scientist Assistant Intern – Image Restoration

Jul. 2023 – Sep. 2023

Chinese Academy of Sciences | TensorFlow, Cloud, GFPGAN

Beijing, China

- Deployed image-restoration modules on Alibaba Cloud via SSH, implementing VAE loss, Gaussian filters, and LSGAN functions for scalable reconstruction workflows.
- Built a TensorFlow 2 CNN on 3.4k train/validation/test images, achieved 80% accuracy across 5 categories, and integrated GFPGAN APIs to reduce processing costs by 5%.